What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side. Did missionary control change *what* they printed or *how* they used words? Imvo was not more political: of ten political topics set in advance, only parliament beats random words, and not after correction. Religion, an exploratory topic, differs most: Isigidimi uses it almost four times as often, also in readers' letters and under the same editor (uncorrected). How political words were used barely differs. Old spelling costs only the African-language tokenizer more. The evidence fits control shaping what the papers printed; any effect on how they wrote is below what we can detect.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? The IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a) lets us ask this. *Isigidimi sama-Xosa* was a missionary paper. *Imvo Zabantsundu* was founded in 1884 by Isigidimi's former editor, J. T. Jabavu, as the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). After 1884 both papers had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71). But comparing the papers also compares two editors, places and readerships, so we also compare Jabavu with himself: he edited Isigidimi from 1881 to 1884 and Imvo afterwards.

Communication research distinguishes what a paper covers, *agenda-setting* (McCombs & Shaw, 1972), from how it presents it, *framing* (Entman, 1993). We measure coverage as **salience**: how often a topic's words appear. We study framing with word embeddings, which place each word by its neighbours (Hamilton et al., 2016a). Most such studies use modern news (Card et al., 2015; Field et al., 2018); Schoots (2023b) counted root words in these papers. We add controlled comparisons, a same-editor test and a letters-versus-rest split, and ask whether this history affects how machines read the archive (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi uses political words in more religious contexts. **H3**: old text and spelling cost tokenizers more.

Methods

Data. We use the archive's human-corrected transcriptions, which were not spell-checked (Schoots et al., 2023a, p. 59), from November 1884 to 1888: 286,663 words of Isigidimi and 1.31 million of Imvo (safeguards in Table 1).

Salience. Each topic is a set of word stems, adapted from Schoots (2023b), including spelling variants such as Imvo's *rulumeni* for *rulumente*, 'government' (Table S1). We fixed ten political topics in advance, plus four neutral controls. Religion (*tixo*, *kristu*) is exploratory: a classifier trained on the same years suggested it (logistic regression, issue-grouped nested cross-validation, 0.83 balanced accuracy; chance 0.50), though the archive's authors also expect it (Schoots et al., 2023a, p. 57). For rates per million words, bootstrap confidence intervals over whole issues test whether the rates differ; matched random word sets ask the stricter question: does it differ more than random words of the same frequency? Jabavu's Isigidimi runs January 1881–October 1884.

Threat	Response
Different years, sizes	Nov 1884–88; ten month- and size-matched draws
Frequency effects	2000 matched random word sets (Monroe et al., 2008)
Change over time	Each paper's own drift in other years
Unstable embeddings	Tune for seed stability; average ten runs
Non-political effects	Four neutral control concepts
One paper per side	Same editor in both; place, audience unfixed

Table 1: Threats to a fair comparison, and our response.

Letters. A reader's letter opens by addressing the editor (*nkosi mhleli . . . kaundifakele*, 'sir, editor . . . please print'). In a hand check, 28 of 30 detected letters were genuine (Table S5).

Framing. Embeddings capture a word's meaning, grammar and topic, so they are only a rough proxy for framing in Entman's sense. We train fastText embeddings (Bojanowski et al., 2017), tuning 34 settings (Table S8) on other years for stability across random seeds, since unstable embeddings create false shifts (Antoniak & Mimno, 2018). This narrow window favours grammar, so we also report wider ones. A word's *shift* measures how differently it relates to 500 common words in each paper (Hamilton et al., 2016b); shuffling issues between the papers gives the level expected by chance (Dubossarsky et al., 2017). A directional test, aimed at framing, asks whether political words sit closer to religious or political words (Table S4). We also count religious and legal words near *umhlaba*.

Tokenizers. We count subword pieces that XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT (Devlin et al., 2019) need for modern news (Adelani et al., 2023), the same news in simulated old spelling, and our text.

Results

H1: different topics, but not more politics. Of the ten political topics, only parliament beats random words: Imvo uses it 3.3 times as often ($p = 0.016$; Fig. 1, left). Imvo uses law, rule and government more too, but not beyond random words; vote, rights and land do not differ. Religion differs most: Isigidimi uses it 3.8 times as often ($p = 0.019$). Neither result survives correction for 15 tests. The religious

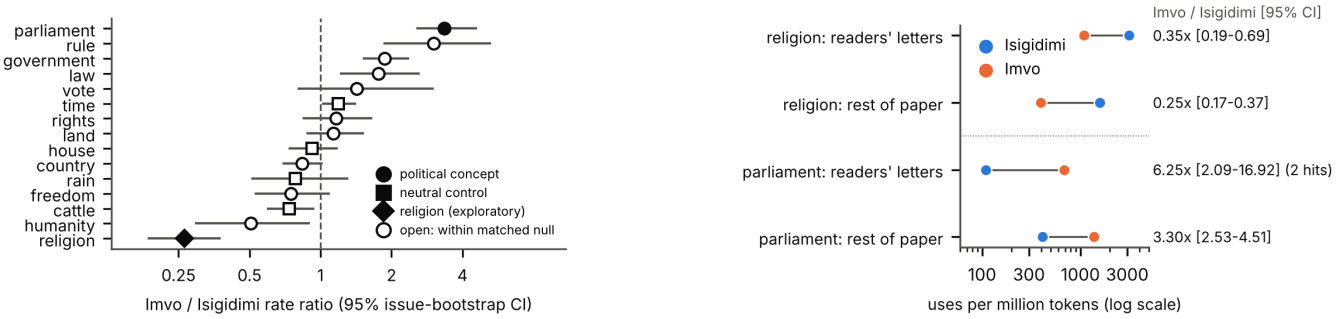


Figure 1: What the papers printed. Left: Imvo's rate relative to Isigidimi's, per topic (95% CI; filled: beats random word sets, $p < 0.05$ uncorrected). Right: the religious gap also appears in letters.

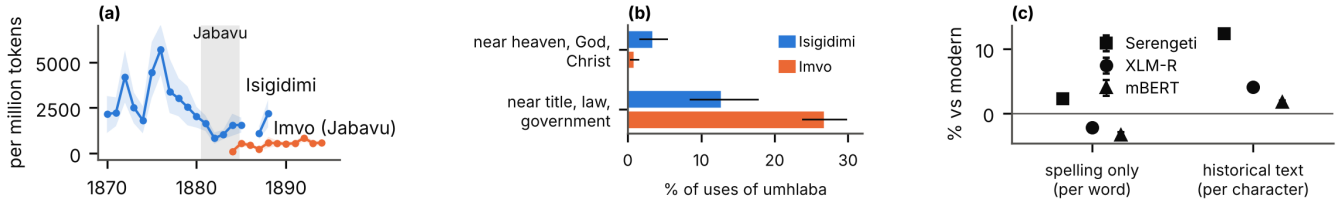


Figure 2: (a) Religious words per million; grey: Jabavu edits Isigidimi, then Imvo; Gqoba edits Isigidimi from late 1884. (b) *Umhlaba* near religious or legal words. (c) Extra subword pieces against modern text (CIs smaller than markers).

gap appears in letters as well as the rest of each paper (Fig. 1, right). Under one editor, Jabavu's Imvo used religious words a third as often as his Isigidimi ($p = 0.025$). Under Gqoba, Isigidimi's rose again, to 1.38 times Jabavu's level (CI 1.03–1.83; $p = 0.168$). Before Imvo existed, Isigidimi's religious words were already falling, while parliament rose 50-fold and vote 12-fold ($p \leq 0.004$; Fig. 2a; Table S7).

H2: little difference in how words are used. Overall word use differs (mean shift 0.348, against 0.230 when issues are shuffled), but no topic stands out: parliament shifts no more than the neutral control *cattle* (Fig. S1). Political words do sit closer to religious words in Isigidimi in all five settings (windows 1 and 2, each with and without joined word forms, and window 5), and in three after correction (Table S2, with neighbour overlap; Gonen et al., 2020). But it weakens without one word, *land* (*umhlaba*), and the neutral control *house* shows the same lean; dropping the anchor *endlu* changes neither. The papers also use *umhlaba* differently: Isigidimi puts it near religious words in 3.4% of uses, Imvo in 0.9% (Fig. 2b). A single word must shift by 0.15 to be detected, more than the whole gap between the papers (0.118).

H3: historical text costs more. Our historical text needs 12.4%, 4.1% and 1.7% more subword pieces per character with Serengeti, XLM-R and mBERT; per word the sign flips, as word division differs. Simulated old spelling raises Serengeti's per-word cost and lowers the others' (Fig. 2c; Table S9).

Discussion

Is it control, or just time? (H1). Religious words were falling in Isigidimi anyway, so we combine two comparisons with different blind spots. Imvo against Gqoba's Isigidimi holds the years fixed; Jabavu's two papers hold the editor fixed. Both show the gap, and the second uses years the classifier never saw, partly replicating the religion finding. If the gap were only the falling trend, Isigidimi would have kept falling after 1884; instead it rose again, so control moved religion more than the change of editor did. Had Jabavu's Imvo been as religious as his Isigidimi, control would be refuted; it was not. The argument rests on these two comparisons agreeing, not on their uncorrected p-values. For Jabavu, leaving brought less religion, not clearly more politics (*vote* rose in his Imvo, but was rising anyway).

The gap is not only the editors' voice, or the sections each paper ran. Isigidimi's letter-writers used religious words three times as often as Imvo's, so each paper drew on, or chose, different writers. One wrote to Isigidimi: *ilizwi alizanga ukuba silipate, lize ukuba lisipate* ('the Word did not come for us to carry it; it came to carry [or rule] us'; 1884; our gloss).

What the embeddings show (H2). The framing signal looks like a difference in meaning, not slant: Isigidimi more often uses *umhlaba* for the earth, Imvo for land under law, which alone would move the word in each model. The detection floor rules out large framing differences, not small ones.

Responsible NLP (H3). History still affects today's tools: written isiXhosa took shape in the mission press (cf. Gilmour, 2007), and its old spelling penalises only the model built for African languages. Serengeti's vocabulary has twice XLM-R's share of *th*, *kh* and *ph* pieces (1.9% vs 0.9%), a possible reason. A modern-spelling copy can help tools but must not replace the original: conversion loses information (old *t* can be *t* or *th*). We describe two papers, not isiXhosa speakers; cultural claims without speakers would be extractive (Bird, 2020).

Limitations and conclusion. Two papers, one period, a missing year (1886), no African-owned paper under Gqoba, an automatic letter split, unreviewed translations. Still, the evidence fits control shaping what the papers printed; any effect on how they wrote is below what we can detect.

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